Wind-Assisted Propulsion on North Sea Routes

Maritime performance analytics for Flettner rotors

Aleksandra Kłos Olga Grigorieva Elen Mouradian Bartosz Maj Hubert Jaczyński

DataWaves Innovation Laboratory

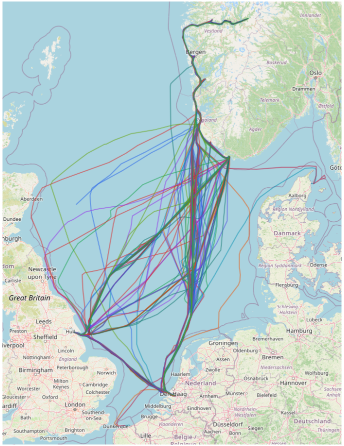
May 2026

Project Question

Aim

Estimate how weather and Flettner rotor assistance affect vessel speed, fuel use, emissions, and route choice.

- Combine AIS trajectories with wind, waves, and currents.
- Predict speed under metocean conditions.
- Convert rotor assistance into speed, fuel, and CO_2 effects.
- Extend the analysis toward uncertainty-aware routing.



Data And Feature Pipeline

- 125,160 AIS-metocean observations.
- 528 voyages from 2023-02-02 to 2026-02-02.
- Chronological voyage split: 422 train voyages, 106 test voyages.
- Features: speed, course, wind angle, wave height, current, and trajectory context.

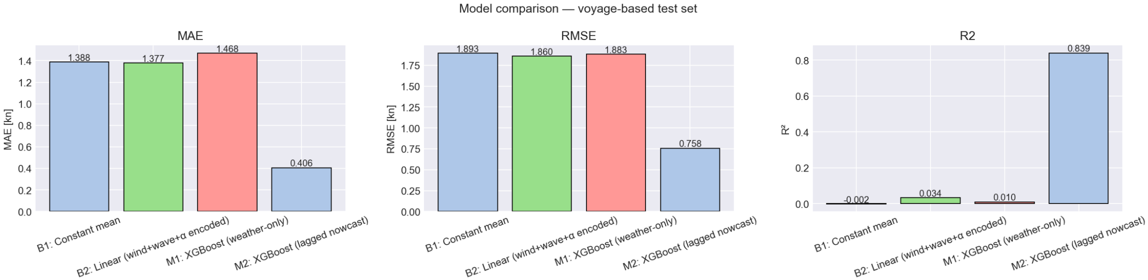
Why the split matters

Observations from one voyage stay on one side of the train/test boundary, reducing temporal and trajectory leakage.

Core target

Speed over ground (SOG), measured in knots.

Speed Model Results



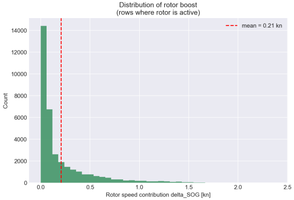
Speed Model Interpretation

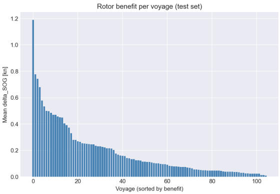
Main result

The strongest model is the lagged nowcast XGBoost model, which uses recent speed and apparent wind context.

- Test MAE: $0.406\,\mathrm{kn}$.
- Test RMSE: $0.758\,\mathrm{kn}$.
- Test R^2: 0.839.
- 91.7% within $\pm 1\,\mathrm{kn}$.
- Weather-only modelling is more useful for interpretation than pure prediction.
- Baselines remain important sanity checks.

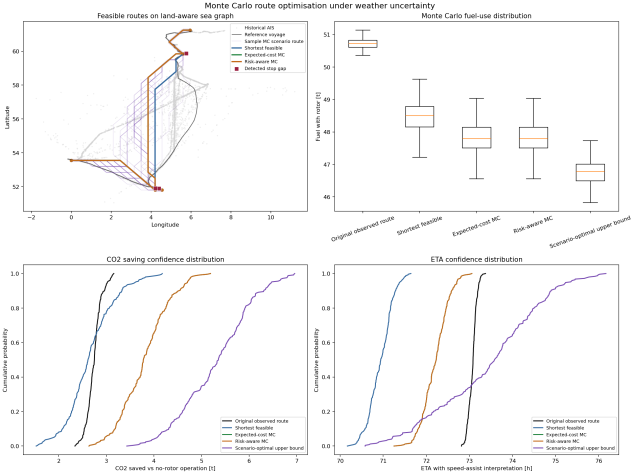
Rotor Opportunity Scenario





- Rotor power is estimated from the provided polar diagram.
- Scenario conversion: $200\,\mathrm{kW} \approx 1\,\mathrm{kn}$ near the operating point.
- Rotor active in 98.1% of held-out observations.
- Mean speed contribution: $0.202\,\mathrm{kn}$ overall.
- Weather-window coverage: 76.0% $\rightarrow$ 77.5%.
- Dataset estimate: 112.2 t fuel and 349.4 t CO_2 avoided.

Monte Carlo Routing Extension



- Keeps the detected itinerary: 4 sailing legs and 3 stop gaps.
- Uses a land-aware graph, with AIS corridor overrides near ports.
- Samples multiple weather and rotor scenarios, not only one optimal path.

Route	nm	fuel	saved	P
Original	863.6	50.72	0.88	–
Shortest	834.1	48.47	0.86	0.00
MC cost	847.9	47.81	1.22	0.86
Scenario best	860.9	46.75	1.70	1.00

Fuel and saved fuel are tonnes; P is probability of beating the shortest feasible route on fuel.

Takeaways And Next Steps

What we have now

- A leakage-aware modelling pipeline.
- A transparent rotor what-if scenario.
- Fuel and CO_2 estimates over voyages.
- A multi-leg, land-aware Monte Carlo routing prototype.
- Route comparisons with uncertainty, confidence intervals, and scenario paths.

See you at the conference!

DataWaves Innovation Laboratory